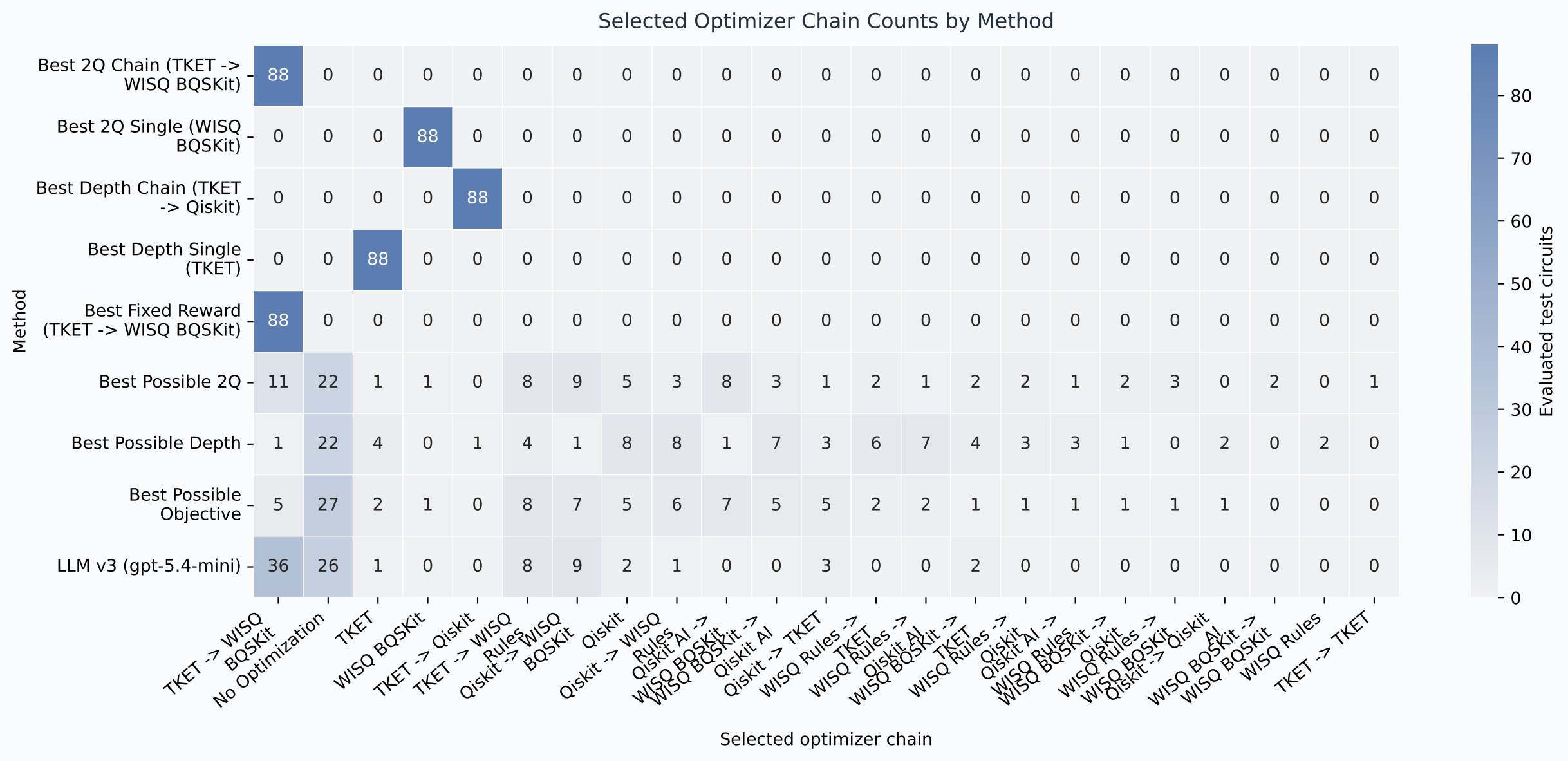
Selected Optimizer Chain Counts by Method
Method
Best 2Q Chain (TKET -> WISQ BQSKit)
Best 2Q Single (WISQ BQSKit)
Best Depth Chain (TKET -> Qiskit)
Best Depth Single (TKET)
Best Fixed Reward (TKET -> WISQ BQSKit)
Best Possible 2Q
Best Possible Depth
Best Possible Objective
LLM v3 (gpt-5.4-mini)
88 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 88 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 0 0 88 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 0 88 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
88 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
11 22 1 1 0 8 9 5 3 8 3 1 2 1 2 2 1 2 3 0 2 0 1
1 22 4 0 1 4 1 8 8 1 7 3 6 7 4 3 3 1 0 2 0 2 0
5 27 2 1 0 8 7 5 6 7 5 5 2 2 1 1 1 1 1 1 0 0 0
36 26 1 0 0 8 9 2 1 0 0 3 0 0 2 0 0 0 0 0 0 0 0
TKET -> WISQ BQSKit
No Optimization
TKET
WISQ BQSKit
TKET -> Qiskit
TKET -> WISQ Rules
Qiskit -> WISQ BQSKit
Qiskit
Qiskit -> WISQ Rules
Qiskit AI -> WISQ BQSKit
Qiskit AI
Qiskit -> TKET
WISQ Rules -> TKET
WISQ Rules -> Qiskit AI
WISQ BQSKit -> Qiskit AI
TKET
WISQ Rules -> Qiskit
Qiskit AI -> WISQ Rules
WISQ BQSKit -> Qiskit
WISQ Rules -> WISQ BQSKit
Qiskit -> Qiskit AI
WISQ BQSKit -> WISQ BQSKit
WISQ Rules
TKET -> TKET
Selected optimizer chain
Evaluated test circuits
0
10
20
30
40
50
60
70
80